UNIT

Two

Behavior of the function and curve sketching

Lesson 1 Increasing and decreasing functions.

Lesson 2 Local maxima and minima (Local extrema).

Lesson 3 Absolute maxima and minima (Absolute extrema).

Lesson 4 Convexity of curves and inflection points.

Lesson 5 Curve studying.

Lesson 6 Applications of maxima and minima.



Unit 2

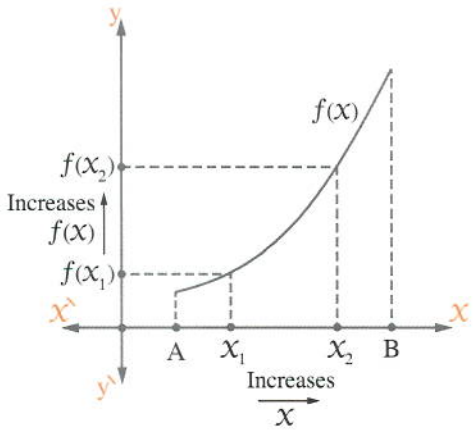
Lesson One

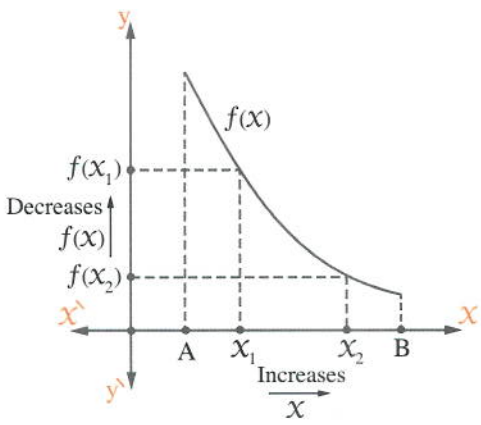
Increasing and decreasing functions

Definition

⊙ If the function f is defined on the interval $]a, b[$, then for every $x_1 < x_2$ in this interval if :

1. $f(x_1) < f(x_2)$, then the function is increasing.
2. $f(x_1) > f(x_2)$, then the function is decreasing.





i.e. The function is increasing on an interval if its value increases as x is increasing and the function is decreasing if its value decreases as x is increasing.

First derivative test for the monotony of the functions

Theorem " without Proof "

1. If $f'(x) > 0$ for all the values of $x \in]a, b[$, then f is increasing on this interval
2. If $f'(x) < 0$ for all the values of $x \in]a, b[$, then f is decreasing on this interval

Remarks

- The function is increasing on an interval if the slope of the tangent to its curve in this interval is positive.

  i.e. The tangent makes an acute angle with the positive direction of X-axis.

- The function is decreasing on an interval of the slope of the tangent to its curve in this interval is negative.

  i.e. The tangent makes an obtuse angle with the positive direction of X-axis.

Definition of the critical point

The continuous function f on the interval $]a , b[$ has a critical point $\left(c , f(c)\right)$
If $c \in]a , b[$ and $f'(c) = 0$ or $f'(c)$ not exist.

For example :

1. If $f(X) = X^3 - 3X$,

then $f'(X) = 3X^2 - 3 = 3(X^2 - 1) = 3(X - 1)(X + 1)$

$\therefore f'(X) = 0$ at $X = 1$, $X = -1$

$\therefore$ The function has a critical points at $X = 1$, $X = -1$

, $\because f(1) = -2$, $f(-1) = 2$

$\therefore$ The critical points are $(1 , -2)$, $(-1 , 2)$

2. If $g(X) = X^{\frac{1}{3}}$, then $g'(X) = \frac{1}{3} X^{\frac{-2}{3}} = \frac{1}{3X^{\frac{2}{3}}}$

$\therefore g'(X)$ is undefined at $X = 0$

$\therefore$ The function has a critical point at $X = 0$

, $\because f(0) = 0$

$\therefore$ The critical points is $(0 , 0)$

Important Remark

The critical point at $X = a$ must be in the domain of the function

i.e. $f(a)$ must be defined.

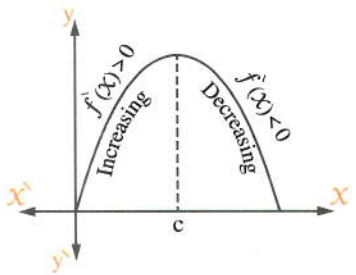
For example :

If $f(x) = \frac{1}{\sqrt{x-1}}$, then $f'(x) = -\frac{1}{2}(x-1)^{\frac{-3}{2}}$

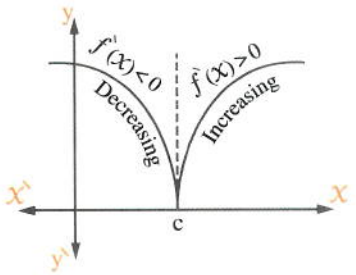
although $f'(1)$ is undefined , the function has no critical point at $x = 1$

because $[f(1)$ is undefined where $x = 1 \notin$ the domain of the function$]$

Steps to determine the increasing and decreasing using the first derivative :



There is a critical point at $x = c$ because $f'(c) = 0$



There is a critical point at $x = c$ because $f'(c)$ undefined.

1. Determine the domain of the function.
2. Find $f'(x)$
3. Find the critical point. (the points at which $f'(x) = 0$ or $f'(x)$ is not exist)
4. Determine the intervals of the domain by which these points divide the domain.
5. Determine the sign of $f'(x)$ in each of these intervals and so the increasing intervals where $[f'(x) > 0]$ and the decreasing intervals where $[f'(x) < 0]$

Example 1

Find the critical points then determine the increasing and decreasing intervals of the function f : $f(x) = x^2 - 4x + 5$

Solution

$\because f(x) = x^2 - 4x + 5$ $\quad\therefore$ The domain of the function is $\mathbb{R}$

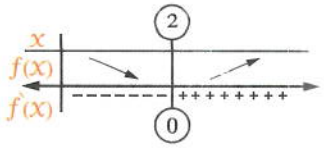
, $f'(x) = 2x - 4$

Put $f'(x) = 0$ $\quad\therefore 2x - 4 = 0$ $\quad\therefore x = 2$

, $\because f(2) = 1$

$\therefore$ The critical point is $(2 , 1)$

$\therefore x = 2$ divides the domain into two intervals.



∴ The function is increasing on

$]2 , \infty[$ and decreasing on $]-\infty , 2[$

To determine the sign of $f`(x)$

Choose a number $\in]-\infty , 2[$

let it be (-1) and find $f`(-1)$

$f`(-1) = -6 \ (< 0)$

∴ $f`(x)$ is negative for every $x \in]-\infty , 2[$

, then choose a number $\in]2 , \infty[$

let it be (3) and find $f`(3)$

$f`(3) = 2 \ (> 0)$

∴ $f`(x)$ is positive for every $x \in]2 , \infty[$

Another Solution

$\because f(x) = x^2 - 4x + 5$

$\therefore f`(x) = 2x - 4$

⊙ To determine the increasing intervals , solve the inequality $f`(x) > 0$

$\therefore 2x - 4 > 0$ $\quad \therefore x > 2$ $\quad \therefore$ The function is increasing on $]2 , \infty[$

⊙ To find the decreasing intervals , solve the inequality $f`(x) < 0$

$\therefore 2x - 4 < 0$ $\quad \therefore x < 2$ $\quad \therefore$ The function is decreasing on $]-\infty , 2[$

Example 2

Find the critical points , then determine the increasing and decreasing intervals of the function $f : f(x) = x^3 - \frac{3}{2}x^2$

Solution

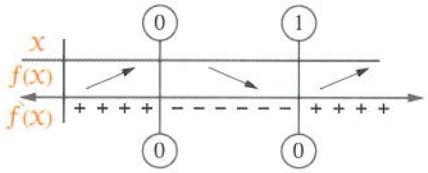
$\because f(x) = x^3 - \frac{3}{2}x^2$ $\quad \therefore$ The domain of the function is $\mathbb{R}$

, $\because f`(x) = 3x^2 - 3x = 3x(x-1)$

Put $f`(x) = 0$ $\quad \therefore 3x(x-1) = 0$ $\quad \therefore x = 0$, $x = 1$

, $\because f(0) = 0$, $f(1) = -\frac{1}{2}$

∴ The critical points are : $(0 , 0)$, $\left(1 , -\frac{1}{2}\right)$

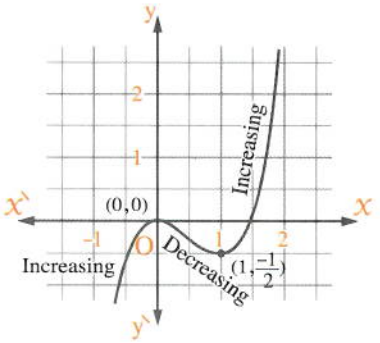


∴ The function is increasing on

the intervals $]-\infty , 0[$, $]1 , \infty[$

and decreasing on $]0 , 1[$

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Remark

The quadratic function $f(x) = a x^2 + b x + c$ has the sign just like sign of a (coefficient of x^2) except if the function has two real roots , then x between these two roots has different sign of a

in the previous example : $f'(x) = 3x(x-1)$

The sign of $f'(x)$ is positive except between $x = 0$, $x = 1$, its sign is negative

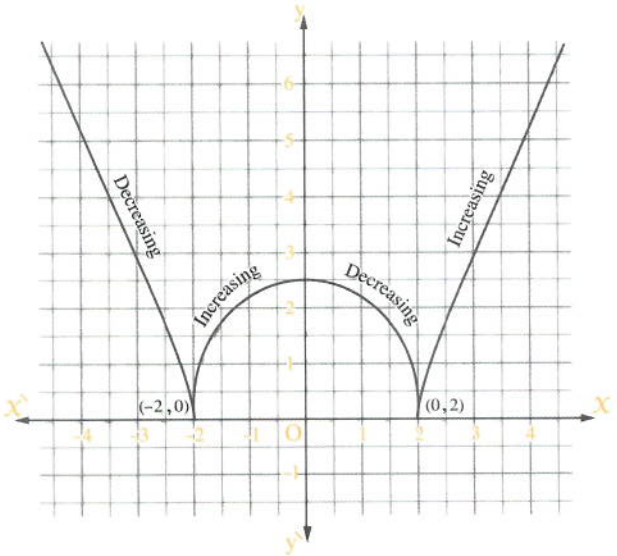
i.e. $f'(x)$ is positive for $x \in]-\infty , 0[\cup]1 , \infty[$ (the function is increasing)

, $f'(x)$ is negative for $x \in]0 , 1[$ (the function is decreasing)

Example 3

Find the critical points , then determine the increasing and decreasing intervals of the function f : $f(x) = (x^2 - 4)^{\frac{2}{3}}$

For reading only



Solution

$\because f(x) = (x^2 - 4)^{\frac{2}{3}}$

$\therefore$ The domain of function is $\mathbb{R}$

, $f'(x) = \frac{2}{3}(x^2 - 4)^{-\frac{1}{3}} \times 2x$

$= \frac{4x}{3(x^2 - 4)^{\frac{1}{3}}}$

$\therefore f'(x) = 0$

At $4x = 0$, $\therefore x = 0$

$\because f(0) = \sqrt[3]{16}$

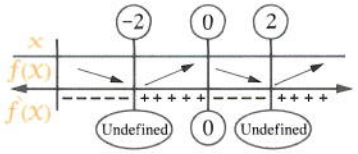
$\therefore$ There is a critical point at $x = 0$, $\left(0 , \sqrt[3]{16}\right)$

, $f'(x)$ is undefined at $x^2 - 4 = 0$

i.e. $(x - 2) = 0$, $(x + 2) = 0$

$\therefore x = 2$, $x = -2$, $\because f(2) = f(-2) = 0$

$\therefore$ There are critical points at $x = 2$ is $(2 , 0)$ and at $x = -2$ is $(-2 , 0)$



∴ The function is increasing on the intervals $]-2\ ,0[\ ,\ ]2\ ,\infty[$ and decreasing on $]-\infty\ ,-2[\ ,\ ]0\ ,2[$

To determine the sign $f'(x)$

Choose a number $\in\ ]-\infty\ ,-2[$

let it be (-3) so

$$f'(-3)=\frac{4(-3)}{3(9-4)^{\frac{1}{3}}}=\frac{\text{negative}}{\text{positive}}=\text{negative}$$

choose a number $\in\ ]-2\ ,0[$

let it be (-1)

So $f'(-1)=\dfrac{4\times-1}{3(1-4)^{\frac{1}{3}}}=\dfrac{\text{negative}}{\text{negative}}=\text{positive}$

and so on ...

Example 4

Find the critical points , then find the increasing and decreasing intervals of the function $f: f(x)=\dfrac{x^4+1}{x^2}$

Solution

$\because f(x)=\dfrac{x^4+1}{x^2}$

∴ The domain of the function is $\mathbb{R}-\{0\}$

$$, f'(x)=\frac{x^2\times 4x^3-(x^4+1)(2x)}{x^4}=\frac{2(x^4-1)}{x^3}$$

$\therefore f'(x)=0$ at $x^4-1=0$ $\quad\therefore x^4=1$ $\quad\therefore x=\pm 1$

, $\because f(1)=f(-1)=2$

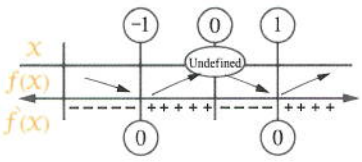
∴ There are critical points at $x=\pm 1$

, they are $(1\ ,2)\ ,(-1\ ,2)$

$f'(x)$ is undefined at $x=0$

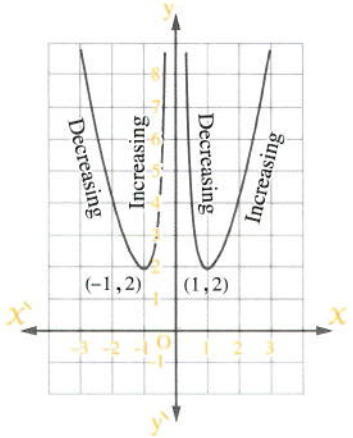
, $\because 0\notin$ the domain of the function

∴ There is no critical point at $x=0$



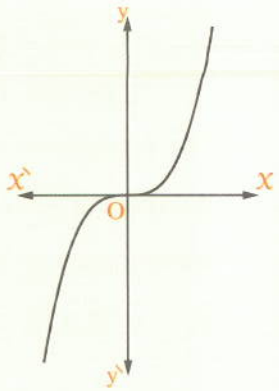
∴ The function is increasing on $]-1\ ,0[\ ,\ ]1\ ,\infty[$ and decreasing on $]-\infty\ ,-1[\ ,\ ]0\ ,1[$

For reading only



Enrich your knowledge

1. The previous theorem verified to the function continuous on the interval $]a , b[$ even if it contains some divergent points where "$f'(x) = 0$" or "$f'(x)$ not exist."



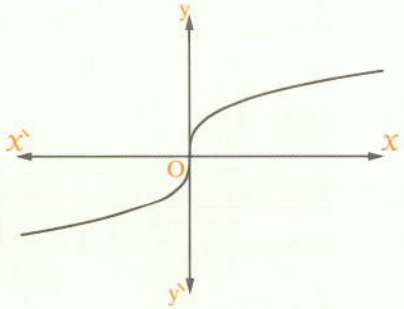
For example :

⊙ The function $f : f(x) = x^3$

from the graph , the function is continuous and increasing on $\mathbb{R}$ even (though) $f'(0) = 0$

So $f'(x) > 0$ for every $x \neq 0$, $f'(x) = 0$ at $x = 0$

⊙ The function $f : f(x) = \sqrt[3]{x}$ from the graph , the function is continuous and increasing on $\mathbb{R}$ even though $f'(0)$ not exist.



i.e. $f'(x) > 0$ for every $x \neq 0$, $f'(x)$ not exist at $x = 0$

and so we can deduce that :

⊙ If f is a continuous function in the interval $]a , b[$, $\{c_1 , c_2 , ..., c_n\} \subset]a , b[$ and $f'(x) > 0$ for every $x \in]a , b[- \{c_1 , c_2 ..., c_n\}$, $f'(x) = 0$ or undefined for every $x \in \{c_1 , c_2 , ..., cn\}$, then the function f is increasing on the interval $]a , b[$

2. If $f'(x) = 0$ for every $x \in]a , b[$, then the function is constant in this interval knowing that: $x \in]a , b[$ and it is meaningless that to say the function is increasing or decreasing at a point.

Example 5

Find the critical points , then determine the increasing and decreasing intervals of the function

$f : f(x) = x^3 - 6x^2 + 12x - 5$

Solution

$\because f(x) = x^3 - 6x^2 + 12x - 5$

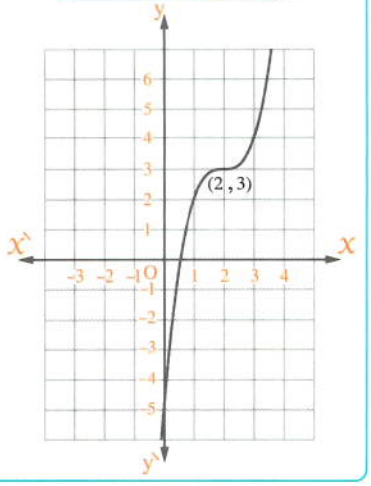
$\therefore$ The domain of the function $f(x)$ is $\mathbb{R}$

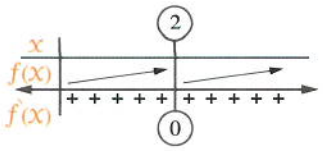
, $f'(x) = 3x^2 - 12x + 12 = 3(x^2 - 4x + 4) = 3(x-2)^2$

put $f'(x) = 0$ $\quad \therefore (x-2)^2 = 0 \quad \therefore x = 2$

, $\because f(2) = 3$ $\quad \therefore$ The critical point is $(2 , 3)$

For reading only





$\therefore$ The function is increasing on $]-\infty , 2[$ and $]2 , \infty[$

, $\because$ the function is continuous at $x = 2$

$\therefore$ The function is increasing on $\mathbb{R}$

Notice that

$f`(x) = 3\,(x-2)^2 > 0$ for all values of $x \in \mathbb{R} - \{2\}$

, $\because$ the function is continuous at $x = 2$ $\therefore$ The function is increasing on $\mathbb{R}$

Example 6

Find the critical points , then determine the increasing and decreasing intervals of the function $f : f(x) = \frac{x}{x-2}$, $x \neq 2$

Solution

$\because f(x) = \frac{x}{x-2}$

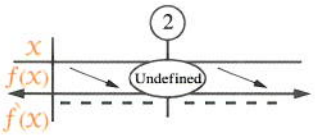
$\therefore$ The domain of the function is $\mathbb{R} - \{2\}$

, $f`(x) = \frac{(x-2)-x}{(x-2)^2} = \frac{-2}{(x-2)^2}$

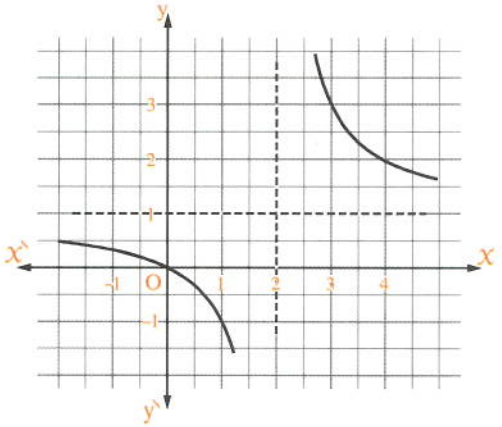
$\therefore f`(x)$ is undefined at $x = 2$

, $\because 2 \notin$ the domain

$\therefore$ There is no critical points to the function



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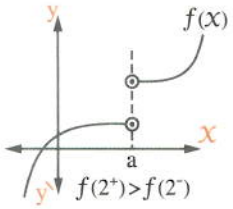
$\therefore$ The function is decreasing on $]-\infty , 2[$, $]2 , \infty[$

Notice that

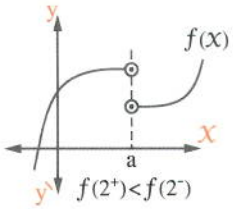
Even though the function is decreasing on each of the two intervals $]-\infty , 2[$ and $]2 , \infty[$ we can not say that it is decreasing on its domain $\mathbb{R} - \{2\}$ because $f(2^+) > f(2^-)$

Remarks

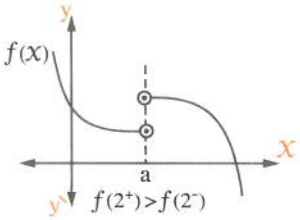
In case the function is not continuous at $\mathcal{X} = a$



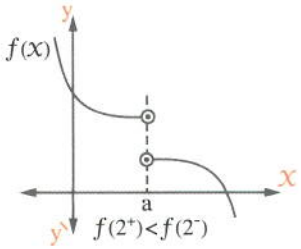
f is increasing on $\mathbb{R} - \{a\}$



f is increasing on $]-\infty , a[\ ,]a , \infty[$



f is decreasing on $]-\infty , a[\ ,]a , \infty[$



f is decreasing on $\mathbb{R} - \{a\}$

Example 7

Determine the increasing and decreasing intervals of the function f :

$f(\mathcal{X}) = \sin \mathcal{X} + \cos \mathcal{X}$ on the interval $]0 , 2\pi[$

Solution

$\because f(\mathcal{X}) = \sin \mathcal{X} + \cos \mathcal{X}$ is continuous on the interval $]0 , 2\pi[$

, $\because \grave{f}(\mathcal{X}) = \cos \mathcal{X} - \sin \mathcal{X}$

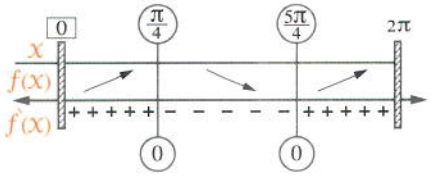
put $\grave{f}(\mathcal{X}) = 0$

$\therefore \cos \mathcal{X} - \sin \mathcal{X} = 0$

$\therefore \cos \mathcal{X} = \sin \mathcal{X}$

$\therefore \tan \mathcal{X} = 1$

$\mathcal{X} = \frac{\pi}{4}$ or $\mathcal{X} = \frac{5\pi}{4}$



$\therefore$ The function is increasing on $]0 , \frac{\pi}{4}[$,

$]\frac{5\pi}{4} , 2\pi[$ and decreasing on $]\frac{\pi}{4} , \frac{5\pi}{4}[$

Example 8

Find the critical points then determine the increasing and decreasing intervals of the function $f : f(x) = e^x - x$

Solution

$\because f(x) = e^x - x$

$\therefore$ The domain of the function is $\mathbb{R}$

, $f'(x) = e^x - 1$

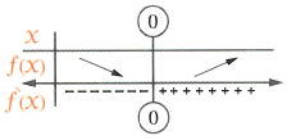
put $f'(x) = 0$ $\quad\quad$ $\therefore e^x - 1 = 0$

$\therefore e^x = 1$ $\quad\quad$ $\therefore x = 0$

$\therefore$ There is a critical point at $x = 0$

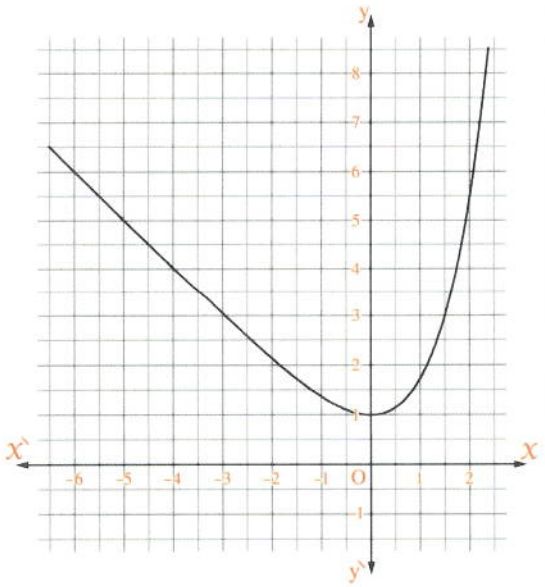
$\because f(0) = 1$

$\therefore$ $(0 , 1)$ is the critical point



$\therefore$ The function is decreasing on $]-\infty , 0[$ and increasing on $]0 , \infty[$

For reading only



Example 9

Find the critical points , then determine the increasing and decreasing intervals of the function $f : f(x) = x^2 - 2 \ln x$

Solution

$\because f(x) = x^2 - 2 \ln x$

$\therefore$ The domain of the function is $\mathbb{R}^+$

, $f'(x) = 2x - \frac{2}{x}$

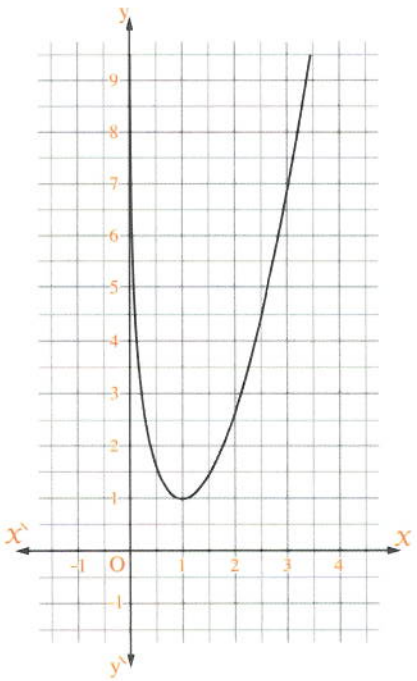
, put $f'(x) = 0$ $\quad\quad$ $\therefore 2x - \frac{2}{x} = 0$

$\therefore 2x = \frac{2}{x}$ $\quad\quad$ $\therefore x^2 = 1$

$\therefore x = 1 , x = -1$ (Refused : $-1 \notin \mathbb{R}^+$)

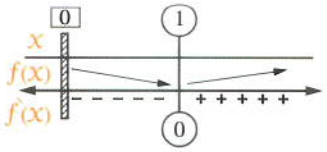
$\therefore$ There is a critical point at $x = 1$

For reading only



$\because f(1) = (1)^2 - 2 \ln 1 = 1$

$\therefore (1 , 1)$ is a critical point.



$\therefore$ The function is decreasing on $]0 , 1[$ and decreasing on $]1 , \infty[$

Example 10

Find the critical points , then determine the increasing and decreasing intervals of the function $f : f(x) = x|x-2|$

Solution

$$\because f(x) = \begin{cases} x(x-2) , x \geq 2 \\ -x(x-2) , x < 2 \end{cases} = \begin{cases} x^2 - 2x , x \geq 2 \\ -x^2 + 2x , x < 2 \end{cases}$$

$\therefore$ The domain of the function is $\mathbb{R}$

(1) By studying differentiability of the function at $x = 2$

$$\because \grave{f}(2^-) = \lim_{h \to 0} \frac{f(2+h) - f(2)}{h} = -2$$

$$, \grave{f}(2^+) = \lim_{h \to 0} \frac{f(2+h) - f(2)}{h} = 2$$

$\therefore \grave{f}(2^+) \neq \grave{f}(2^-)$ $\quad\quad$ $\therefore \grave{f}(2)$ does not exist

$\because f(2) = 0$ $\quad\quad$ $\therefore (2 , 0)$ is a critical point

$$, \because \grave{f}(x) = \begin{cases} 2x - 2 & , x > 2 \\ \text{not exist} & , x = 2 \\ -2x + 2 & , x < 2 \end{cases}$$

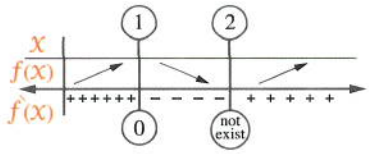
(2) Put $\grave{f}(x) = 0$

⊙ At $x > 2$: $\quad\quad$ $\therefore 2x - 2 = 0$ $\quad\quad$ $\therefore x = 1$ (Refused)

⊙ At $x < 2$: $\quad\quad$ $\therefore -2x + 2 = 0$ $\quad\quad$ $\therefore x = 1$

$\therefore$ There exist a critical point at $x = 1$

, $\because f(1) = 1$



$\therefore$ (1 , 1) is a critical point

$\therefore$ The function is increasing on $]-\infty , 1[$, $]2 , \infty[$
and decreasing on $]1 , 2[$

Remark

We know from the definition of critical point it has to be studied in an open interval , but in some problems there are terminal points at the terminals of a closed interval at which $\dot{f}(x)$ does not exist , these points are called "critical terminal points" but these are out of our syllabus

For example :

If $f(x) = \sqrt{4-x^2}$ $\quad \therefore \dot{f}(x) = \dfrac{-2x}{2\sqrt{4-x^2}} = \dfrac{-x}{\sqrt{4-x^2}}$

, at $\dot{f}(x) = 0$ $\quad \therefore x = 0$ $\quad \therefore$ It is a critical point at $x = 0$

, $\dot{f}(x)$ does not exist at $x = \pm 2$

$\therefore$ There are two critical terminal points at $x = \pm 2$